

NETRA AI: Network Attack Forecasting System

Team Name:- CHILLI BOMB

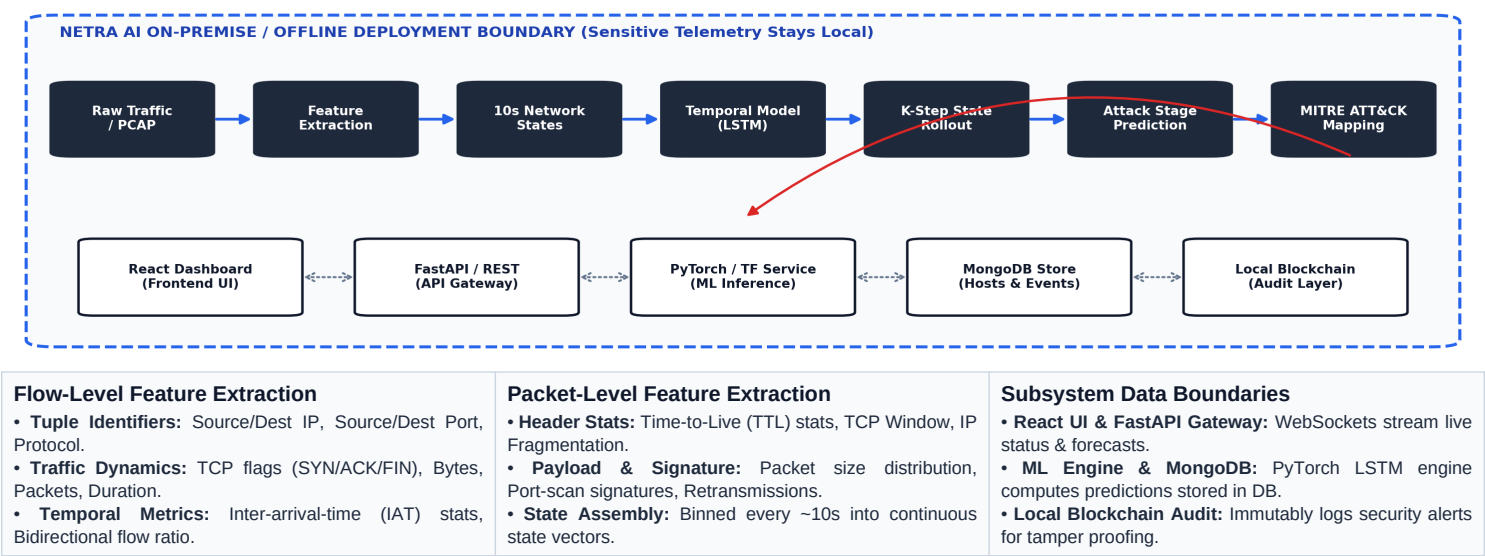
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System Architecture & Core Pipeline Specifications

1. EXECUTIVE SUMMARY & CORE PARADIGM SHIFT

Core Objective: Traditional Network Intrusion Detection Systems (NIDS) and SIEM solutions operate reactively by flagging malicious activity during or after execution. NETRA AI introduces a paradigm shift from reactive detection to predictive network attack forecasting . By continuously ingesting raw network traffic, extracting multi-layered features, and constructing temporal network-state representations, NETRA AI models attack progression dynamics using a World/Temporal Model (LSTM). It forecasts probable next-stage cyberattacks, maps behaviors to MITRE ATT&CK; techniques, and delivers explainable early warnings to analysts prior to malicious impact escalation.	
10-Second Network-State Windowing Groups flow-level (IPs, ports, protocols, flags, bytes, IAT stats) and packet-level (TTL, TCP window, fragmentation, retransmissions) telemetry into contiguous 10s state vectors to capture evolving temporal behavior.	Temporal World Model & K-Step Rollout Employs an LSTM sequence model to learn state-to-state transition probabilities. Executes K-step future state rollouts to simulate future network conditions and calculate estimated attack stage confidence scores.

2. END-TO-END PIPELINE & SUBSYSTEM ARCHITECTURE



3. ATTACK STAGE PROGRESSION SPECTRUM



Stage Transition & Forecasting Mechanics NETRA AI models network state trajectories across 8 discrete lifecycle stages: 1. Normal → 2. Reconnaissance → 3. Initial Access → 4. Persistence → 5. Privilege Escalation → 6. Lateral Movement → 7. Command & Control (C2) → 8. Exfiltration . Rather than issuing isolated static alerts, the temporal forecasting engine computes state transition probabilities across 10s sequence windows, estimating estimated attack stages and confidence scores before malicious impact.	MITRE ATT&CK; Mapping & Threat Intelligence Predicted network state anomalies are automatically mapped to corresponding MITRE ATT&CK; tactics and techniques (e.g., T1046 Network Service Discovery, T1071 Application Layer Protocol, T1041 Exfiltration Over C2 Channel). This provides security analysts with instant tactical context aligned with standard threat frameworks.
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4. EXPLAINABLE RISK INTELLIGENCE & AUDIT INTEGRITY

Analyst Explainability & Evidence Attribution To eliminate 'black-box' ML distrust, NETRA AI extracts key feature contributions (e.g., sudden spikes in SYN flags, atypical TTL distributions, or abnormal bidirectional flow ratios) that led to the forecast. SOC analysts receive actionable evidence explaining exactly <i>why</i> a future attack stage is estimated.	Cryptographic Blockchain Audit Trail Includes a local, tamper-evident blockchain ledger dedicated to recording security alerts, forecast outputs, and model decision logs. This guarantees data integrity, prevents historical log tampering, and fulfills strict compliance and forensic traceability requirements in controlled SOC environments.
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Component Specifications, Deployment & Roadmap

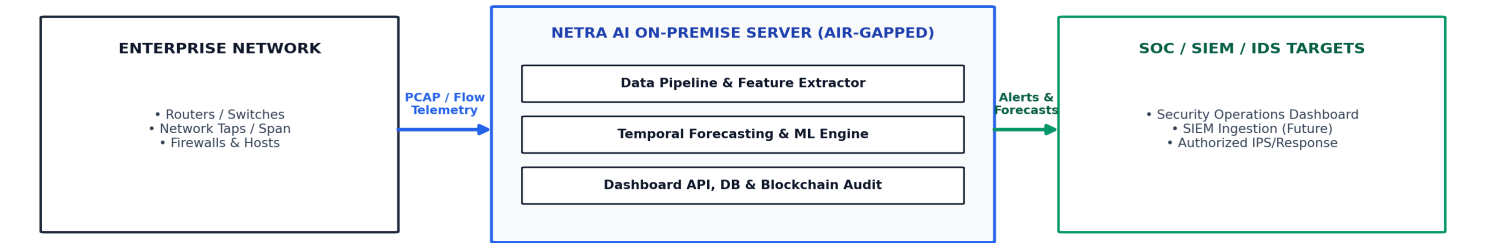
5. SUBSYSTEM COMPONENT ARCHITECTURE MATRIX

Subsystem Component	Primary Functionality & Role	Input Telemetry / Data	Output Artifact / Stream
Frontend Web Dashboard	React-based SOC interface displaying live threat status, network graph, traffic feeds, forecasts & risk analysis.	WebSocket streams, REST API JSON	Analyst UI views & controls
Backend API Gateway	FastAPI server orchestrating data pipelines, model inference requests, live WebSockets & database transactions.	HTTP/REST requests, Socket.IO feeds	JSON API endpoints & streaming state
ML Forecasting Engine	PyTorch/TensorFlow temporal model executing LSTM state transitions & K-step future rollout simulations.	10s state sequence tensors	Predicted state, stage & confidence
Data Ingestion Pipeline	Extracts flow and packet features from raw PCAP/telemetry and groups them into 10s state windows.	Raw PCAP / Flow telemetry data	Structured 10s state feature vectors
Database Store	MongoDB database storing persistent host information, traffic events, forecast histories & alert logs.	Host data, events & forecast outputs	Persisted documents & query results
MITRE ATT&CK; Engine	Maps forecasted network state features to standard MITRE tactics, techniques, and threat categories.	Predicted state feature vectors	MITRE Technique IDs & labels
Explainability Engine	Computes feature attributions and evidence indicators supporting estimated future attack stages.	Model weights & input features	Feature importance & evidence panel
Audit & Integrity Layer	Local blockchain ledger ensuring tamper-evident, verifiable recording of security alerts and forecast events.	Alerts, forecasts & audit records	Cryptographic block hashes & logs
Integration Layer	Modular API architecture enabling seamless hooks into enterprise SIEM, IDS/IPS & SOC monitoring systems.	Normalized alert & forecast feeds	SIEM events & action triggers

6. END-TO-END TELEMETRY PROCESSING SEQUENCE

1. Ingestion: Capture PCAP / NetFlow.	2. Extraction: Compute flow & packet features.	3. Windowing: Bin features into 10s states.	4. Sequencing: Form temporal state series.
5. Model Inference: LSTM state transition forecasting.	6. Estimation: Estimate stage & confidence.	7. MITRE Mapping: Map to ATT&CK; techniques.	8. Alerting: Push WebSockets alert to SOC UI.

7. ON-PREMISE AIR-GAPPED DEPLOYMENT ARCHITECTURE



Data Privacy & Local Telemetry Isolation Network traffic and security telemetry contain highly sensitive IP addresses, hostnames, payload signatures, and operational patterns. Local processing guarantees that sensitive telemetry never leaves the organization's secure infrastructure boundary.	Zero Cloud Dependency & Air-Gapped Security Eliminates reliance on external cloud APIs, internet connectivity, or third-party infrastructure. Ensures operational resilience, low latency, and full compliance with strict high-security, controlled SOC environments.	Integrity Assurance & Blockchain Auditability Local storage paired with local blockchain audit logging fulfills stringent government and defense compliance requirements, providing immutable proof of alert delivery without external data leakage.
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8. TECH STACK, DESIGN PRINCIPLES & PROJECT ROADMAP

Verified Project Technology Stack Frontend: React.js, Network Graph Visualizer, Socket.IO Client. Backend & ML: Python 3.x, FastAPI, PyTorch / TensorFlow (LSTM). Database & Audit: MongoDB, Local Blockchain Ledger. Demonstration Dataset: CTU-13 Neris Botnet PCAP traffic.	Core Security & Architectural Principles On-Premise / Offline Processing: Keeps telemetry inside enterprise boundary. Explainable Predictions: Feature attribution eliminates black-box ML. Model Confidence & Rollout: K-step state simulation with probability bounds. Auditability & Modular APIs: Blockchain logging & integration-ready endpoints.
Current Implementation Baseline (Verified Scope) - Local PCAP and network traffic file analysis. - Feature extraction & 10-second network-state generation. - Temporal LSTM forecasting & attack-stage estimation. - MITRE ATT&CK; mapping & explainability evidence UI. - React web dashboard with forecast history & live updates. - Local blockchain verification for tamper-evident audit logs. - Tested and verified using CTU-13 Neris botnet PCAP dataset.	Future Enterprise Integration Targets (Roadmap) - Direct enterprise SIEM system integration (e.g., Splunk, QRadar). - Continuous live network tap telemetry ingestion. - IDS/IPS integration for automated block/mitigation responses. - Full SOC workflow orchestration and incident management. - Multi-subnet scalable enterprise cluster deployment.